

Finite harmonic expansion of the evaluated-phase residual flux

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1 Scope and chart

This memo expands the evaluated-phase coefficient already defined in `moments.body.tex` and the finite chart of `quantitative.extension.tex`. It is a fixed-band, finite-label, compact-support statement. The physical viscosity is the positive number ν_{NS} ; it is not replaced by ε or by the source clock $\nu = \sqrt{A_0}$. Put

$$0 < Q \leq 1, \quad A = \frac{1}{2} + h, \quad D = \frac{1}{2} - h, \quad \varepsilon = Q^h, \quad r = \sqrt{Q}R, \quad z = Q^D Z.$$

The source constants A_0, λ_0 and $\mu = \lambda_0$ retain their given values. Every statement below is at one such Q and on the strict interior compact patch where all retained coefficients and their displayed Z -derivatives are smooth and compactly supported in R .

2 Exact angular kernel

Let I_* and I_y be the finite retained old-parent and new-harmonic label sets. For $\tau \in I_*$ and $j \in I_y$, let $n_\tau, n_j \in \mathbb{Z} \setminus \{0\}$ be their actual angular carriers. Write their cylindrical component coefficients, before taking real parts, as

$$W_{\tau,\ell} = \Re(a_{\tau,\ell} e^{in_\tau \theta}), \quad V_{j,\ell} = Q^{-A} \Re(\zeta_{j,\ell} e^{in_j \theta}), \quad \zeta_{j,\ell} = b_{j,\ell} e^{i\varphi_j},$$

where $\ell \in \{r, \theta, z\}$ and $\varphi_j = k_{\beta_j} m_j \Phi_j$. The old coefficient $a_{\tau,\ell}$ includes its full retained old phase and all cutoff/path factors, so its Z derivative is the full old axial derivative in this representation. Thus

$$\partial_Z \zeta_{j,\ell} = \delta_{j,\ell} e^{i\varphi_j}, \quad \delta_{j,\ell} := \partial_Z b_{j,\ell} + ik_{\beta_j} m_j (\partial_Z \Phi_j) b_{j,\ell}.$$

For complex x, y and integers $n, m \neq 0$ define

$$\mathcal{A}_{n,m}(x, y) := \frac{1}{4} [\mathbf{1}_{n+m=0}(xy + \bar{x}\bar{y}) + \mathbf{1}_{n-m=0}(x\bar{y} + \bar{x}y)]. \quad (\text{E1})$$

Expanding the two real parts and integrating each exponential over $[0, 2\pi]$ proves exactly

$$\langle \Re(xe^{in\theta}) \Re(ye^{im\theta}) \rangle_\theta = \mathcal{A}_{n,m}(x, y). \quad (\text{E2})$$

No equal-frequency assumption is made: the indicators in (E1) retain both $n = m$ and $n = -m$ possibilities. Moreover, for real Z , differentiation commutes with the kernel,

$$\partial_Z \mathcal{A}_{n,m}(x, y) = \mathcal{A}_{n,m}(\partial_Z x, y) + \mathcal{A}_{n,m}(x, \partial_Z y), \quad (\text{E3})$$

because complex conjugation and the integer indicators commute with ∂_Z .

3 Finite stress expansion

Let $\Pi_Y f = \int_{\mathbb{T}^2} f dY$ and $R_Y = I - \Pi_Y$. The stress in the moments calculation is $\mathcal{S} = \langle w_* \otimes v + v \otimes w_* + v \otimes v \rangle_\theta$, and $\mathcal{S}^\circ = R_Y \mathcal{S}$. Since $S^* = Q^{2A} S$, the exact z, τ_e component is

$$\begin{aligned} (S_{z,\tau_e}^\circ)^* = R_Y \Bigg\{ & Q^A \sum_{\tau \in I_*} \sum_{j \in I_y} [\mathcal{A}_{n_\tau, n_j}(a_{\tau,z}, \zeta_{j,\tau_e}) + \mathcal{A}_{n_j, n_\tau}(\zeta_{j,z}, a_{\tau,\tau_e})] \\ & + \sum_{i,j \in I_y} \mathcal{A}_{n_i, n_j}(\zeta_{i,z}, \zeta_{j,\tau_e}) \Bigg\}. \end{aligned} \quad (\text{E4})$$

The first line is every old–new and new–old product; the second is the complete new quadratic sum, including both ordered pairs. Terms whose angular carriers do not satisfy an indicator in (E1) vanish under the angular average, while their labels remain available for the finite sum. The Y -projection is applied after the angular product, so $\langle \mathcal{S}^\circ \rangle_Y = 0$ is retained without cancelling the physical evaluation.

Applying (E3) gives the exact differentiated coefficient. Define

$$\mathcal{D}_{n,m}(x, y) := \mathcal{A}_{n,m}(\partial_Z x, y) + \mathcal{A}_{n,m}(x, \partial_Z y).$$

Then

$$\begin{aligned} \partial_Z(S_{z,\tau_e}^\circ)^* = R_Y \Bigg\{ & Q^A \sum_{\tau,j} [\mathcal{D}_{n_\tau,n_j}(a_{\tau,z}, \zeta_{j,\tau_e}) + \mathcal{D}_{n_j,n_\tau}(\zeta_{j,z}, a_{\tau,\tau_e})] \\ & + \sum_{i,j} \mathcal{D}_{n_i,n_j}(\zeta_{i,z}, \zeta_{j,\tau_e}) \Bigg\}, \end{aligned} \quad (\text{E5})$$

where all sums have the finite ranges $\tau \in I_*$ and $i, j \in I_y$. In (E5), the old terms contain $\partial_Z a_{\tau,\ell}$ and the new terms contain $\partial_Z \zeta_{j,\ell} = \delta_{j,\ell} e^{i\varphi_j}$; no parent derivative has been identified with a new derivative.

4 The coefficient C_e° and its finite envelope

For $e \in \{1, 2\}$, set $\tau_2 = \theta$ and $\tau_1 = z$, and define exactly

$$C_e^\circ(Z, T) := \int_0^\infty R^e \sup_{Y \in \mathbb{T}^2} |\partial_Z(S_{z,\tau_e}^\circ)^*(R, Z, T, Y)| dR. \quad (\text{E6})$$

Because $\|R_Y f\|_{L_Y^\infty} \leq 2\|f\|_{L_Y^\infty}$, the finite sums in (E5) give the explicit envelope

$$\begin{aligned} C_e^\circ \leq 2Q^A \sum_{\tau,j} \int_0^\infty R^e \sup_Y (|\mathcal{D}_{n_\tau,n_j}(a_{\tau,z}, \zeta_{j,\tau_e})| + |\mathcal{D}_{n_j,n_\tau}(\zeta_{j,z}, a_{\tau,\tau_e})|) dR \\ + 2 \sum_{i,j} \int_0^\infty R^e \sup_Y |\mathcal{D}_{n_i,n_j}(\zeta_{i,z}, \zeta_{j,\tau_e})| dR. \end{aligned} \quad (\text{E6a})$$

The indicators in $\mathcal{D}$ make every nonmatching pair exactly zero; therefore (E6a) is a finite pair sum, not an infinite or continuum estimate. The Y -evaluation map $Y = Y(r, t)$ is independent of z , hence the chain rule in the moments body gives

$$M_e^{\text{ev}} - M_e = \partial_z \int_0^\infty r^e \mathcal{S}_{z,\tau_e}^\circ(r, z, t, Y(r, t)) dr, \quad |M_e^{\text{ev}} - M_e| \leq Q^{(e+1)/2-2A-D} C_e^\circ = Q^{e/2-2A+h} C_e^\circ. \quad (\text{E7})$$

The exponent uses $1/2 - D = h$ exactly.

For a directly checkable finite majorant, let

$$\|f\|_{e,\infty Y} := \left(\int_0^\infty R^e \sup_Y |f(R, Y)|^2 dR \right)^{1/2}, \quad \alpha_e := \frac{e+1}{4} - A,$$

and use the finite chart quantities

$$\|a_{\tau,\ell}\|_{e,\infty Y} \leq Q^{\omega_{e,\tau}-(e+1)/4} W_{e,\tau,\ell}, \quad \|\partial_z a_{\tau,\ell}\|_{e,\infty Y} \leq Q^{\dot{\omega}_{e,\tau}-(e+1)/4} \dot{W}_{e,\tau,\ell}, \quad \|b_{j,\ell}\|_e \leq B_{j,e,\ell}, \quad \|\delta_{j,\ell}\|_e \leq D_{j,e,\ell}. \quad (\text{E8})$$

These are finite compact-patch constants; the displayed chart powers include the physical-to-chart measure factor and do not assert uniformity in Q . Since $\partial_Z a = Q^D \partial_z a$, Cauchy–Schwarz and $\|R_Y f\|_\infty \leq 2\|f\|_\infty$ give the old–new contributions to the right side of (E7) with powers

$$Q^{\alpha_e + \dot{\omega}_{e,\tau}} \dot{W}_{e,\tau,\ell} B_{j,e,k}, \quad Q^{\alpha_e + \omega_{e,\tau} - D} W_{e,\tau,\ell} D_{j,e,k}, \quad (\text{E9})$$

for every angularly matched pair and the appropriate components (ℓ, k) . The two exponents are distinct and are not merged unless the actual parent proves $\dot{\omega}_{e,\tau} = \omega_{e,\tau} - D$. The quadratic evaluated terms carry the separate exact power

$$Q^{(e+1)/2-2A-D} = Q^{e/2-2A+h} \quad (\text{E10})$$

against the finite products $D_{i,e,z} B_{j,e,\tau_e}$ and $B_{i,e,z} D_{j,e,\tau_e}$. Thus evaluation preserves the same finite Q powers as the moment rows after the explicit C_e° factor, while retaining all harmonic matching and all old-parent derivative orders.

5 Support, signs, and viscosity

Every term in (E4) contains at least one compactly supported new factor, so the R integrals in (E6) and the boundary terms in the moments are finite and the radial boundary contribution vanishes. The kernel is real and retains the signs of the signed amplitudes and phases. Full reflection reverses the corresponding signed cross flux as dictated by the reflected vector field; absolute envelopes use the same finite constants. No sign of Q , ε , A , D , or ν_{NS} is changed. The retained viscous term remains $-\nu_{\text{NS}}\Delta v$ with its radial, axial, angular, and cylindrical coupling costs. Nothing in the evaluation map supplies a source endpoint or a uniform $Q \downarrow 0$ estimate.

6 Finite conclusion

Equations (E4)–(E10) are the strongest finite evaluated-phase expansion established here: an exact finite harmonic identity, its exact Z derivative, and a finite compact-patch envelope with separate old field and old derivative orders. They do not prove the released third shooting return, a uniform infinite modified sequence, or a smooth terminal-force extension.